The Great Pumpkin Snatch

Whodunnit? · Exponents · Teacher Guide and Answer Key

Unit	Time	Supplies	Group size
Exponents	20–40 minutes	Pencil, scrap paper, scissors	Teams of 2–3 (or solo)

Skills practiced: Exponent rules, zero and negative exponents, exponential equations and models.

General procedures

1. Give each student the **student sheet** (story and case board) and the **Show Your Work** page.
2. Print the **Clue Cards** (pages 3–7 of the student packet) and cut them apart on the dashed lines. Laminating them lets you reuse the set. Post them around the room.
3. Students visit the cards in any order. At each card they solve the problem on their work page, find their answer among the four choices, read the clue beside it, and cross things off the case board. **Every wrong choice leads to a false clue** that clears the real culprit, so a mistake shows up when the board does not narrow down to one suspect, one place and one method.
4. When a team thinks it knows, they bring you the three names. If they are wrong, let them hunt for the card where something went wrong.

Solution: Mr. Corbin took it, in the Parking Field, using this method: Hidden under a tarp.

The reveal (read aloud): While everyone lined up for cider, **Mr. Corbin** backed his truck into the **Parking Field**, hid **Big Orange** under a **tarp**, and drove away as if hauling hay. Every alibi and method you cleared led right to him. A-gourd-able detective work!

Clue map: what each correct clue clears

Card	Clears
1	Suspect: Farmer Whit · Place: Pumpkin Patch
2	Method: Rolled downhill
3	Suspect: Mrs. Alder · Place: Judging Stage
4	Method: Loaded on a hay wagon
5	Suspect: Dottie Pike · Place: Cider Barn
6	Method: Hauled by tractor
7	Suspect: Old Pete · Place: Hay Maze
8	Method: Carted in a wheelbarrow
9	Suspect: Ms. Hale · Place: Petting Zoo
10	Method: Dressed as a scarecrow

Answer Key

The Great Pumpkin Snatch · Correct choice and worked solution for each card

#	Problem	Correct choice	Worked solution	Clue it leads to
1	Simplify: $x^4 \cdot x^3$	**A.** x^7	add the exponents: 4 + 3 = 7 x^7	Farmer Whit was weeding the Pumpkin Patch. Three neighbors waved to him.
2	Simplify: $\frac{x^8}{x^3}$	**D.** x^5	subtract the exponents: 8 − 3 = 5 x^5	The hill behind the patch is untouched. No pumpkin rolled down it.
3	Simplify: $(x^3)^4$	**B.** x^{12}	multiply the exponents: 3 · 4 = 12 x^{12}	Mrs. Alder was announcing winners from the Judging Stage on the loudspeaker.
4	Simplify: $(2x^3)^2$	**A.** $4x^6$	square the 2 and multiply the exponents $4x^6$	The hay wagon was empty. It only held bales all day.
5	Simplify: $3x^2 \cdot 4x^3$	**A.** $12x^5$	multiply coefficients: 3 · 4 = 12; add exponents: 2 + 3 = 5 $12x^5$	Dottie Pike was tasting pumpkin pies in the Cider Barn. The judge's card has the time.
6	Simplify: $\frac{18x^7}{6x^2}$	**B.** $3x^5$	divide 18 by 6 = 3; subtract exponents: 7 − 2 = 5 $3x^5$	The tractor never started. Its engine log shows it parked all evening.
7	Write with a positive exponent: x^{-3}	**D.** $\frac{1}{x^3}$	a negative exponent means reciprocal $\frac{1}{x^3}$	Old Pete was stuffing scarecrows in the Hay Maze with a group of kids.
8	Simplify: $5x^0$ $(x \neq 0)$	**D.** 5	anything (not 0) to the power 0 is 1; 5 · 1 = 5 5	The wheelbarrow axle is fine and not bent. It never held 400 pounds.
9	Solve for x: $3^x = 81$	**C.** $x = 4$	$81 = 3 \cdot 3 \cdot 3 \cdot 3 = 3^4$ $x = 4$	Ms. Hale was selling cider at the Petting Zoo. The goats got the leftovers.
10	An account starts at $500 and grows 20% each year. Write the exponential function for the balance y after x years.	**C.** $y = 500(1.2)^x$	growth factor = 1 + 0.20 = 1.2 $y = 500(1.2)^x$	The scarecrows at the fair all have straw bellies. None hides a pumpkin.

Wrong-Answer Map

The Great Pumpkin Snatch · What a wrong choice looks like, and the false clue it leads to

Card	Wrong choice	Likely error	False clue
1	B. x	subtracted exponents	Farmer Whit was directing trucks in the Parking Field. His orange vest was seen by a dozen drivers.
1	C. x^{12}	multiplied exponents	Mr. Corbin was sipping hot cider in the Cider Barn. The cashier remembers his big tip.
1	D. $2x^{7}$	added the bases	Every tarp at the fair was lifted and checked. Each one covered only folding chairs.
2	A. x^{-5}	subtracted in wrong order	The blue tarp by the barn is soaked and wrinkled flat. Nothing big has been under it.
2	B. x^{24}	multiplied exponents	Ms. Hale was selling cider from a cart in the Parking Field. Her sales slip shows the time.
2	C. x^{11}	added exponents	Mr. Corbin was feeding carrots to the goats in the Petting Zoo. A little girl took his picture.
3	A. x^{81}	raised 3 to the 4th	Dottie Pike was handing out pie samples in the Parking Field. The line of hungry visitors vouches for her.
3	C. x^{7}	added exponents	Mr. Corbin was posing with a giant squash in the Pumpkin Patch. The fair photographer has the shot.
3	D. x	subtracted exponents	The tarp from the tool shed was folded neatly with its corners tucked. It hasn't been moved.
4	B. $2x^{6}$	did not square the coefficient	Mr. Corbin was clapping for the winners at the Judging Stage. The front row saw him the whole time.
4	C. $4x^{9}$	cubed or mis-powered the exponent	Tarp inspectors found no lumps, bumps or pumpkin-shaped shapes anywhere on the grounds.
4	D. $4x^{5}$	added exponents	Old Pete was fixing a flat tire in the Parking Field. The tire shop's receipt has the time.
5	B. $12x^{6}$	multiplied exponents	Mr. Corbin was helping lost kids out of the Hay Maze. The kids still talk about him.
5	C. $7x^{5}$	added coefficients	Mrs. Alder was handing out ribbons in the Parking Field. The ribbon box count backs her up.
5	D. $12x$	subtracted exponents	The big green tarp has no pumpkin stains on it. It only ever covered the hay bales.
6	A. $3x^{9}$	added exponents	Mr. Corbin was refilling the cider jugs in the Cider Barn. The jug count matches the schedule.
6	C. $3x^{14}$	multiplied exponents	The tarp over the tractor was lifted for the inspector. Only an old toolbox was underneath.
6	D. $12x^{5}$	subtracted coefficients	Old Pete was patching a scarecrow in the Parking Field. Stray straw on the ground proves it.
7	A. x^{3}	dropped the negative sign	Mr. Corbin was tying ribbons on vines in the Pumpkin Patch. The ribbon spool is almost empty.
7	B. $-\frac{1}{x^{3}}$	kept an extra negative	A search of the tarps found nothing under them but puddles and a lost mitten.
7	C. $-x^{3}$	made the base negative	Farmer Whit was handing out maps in the Parking Field. The gate attendant wrote down the time.
8	A. -5	applied a negative sign to the coefficient	The tarp rope knots are still tied the way the crew left them. Nobody has untied them.

8	B. 1	forgot the coefficient 5	Mrs. Alder was greeting visitors in the Parking Field. The guest book has her handwriting.
8	C. 0	treated x^0 as 0	Mr. Corbin was braiding a pony's mane in the Petting Zoo. The zoo keeper signed him in.
9	A. x = 9	subtracted 3	Ms. Hale was loading apple crates into a truck in the Parking Field. The driver kept her receipt.
9	B. x = 3	confused 3^3=81	The lumpy tarp turned out to cover a pile of firewood.
9	D. x = 27	divided 81 by 3	Mr. Corbin was fixing a speaker cable at the Judging Stage. The sound crew watched him work.
10	A. y = 500(1.2x)	multiplied by x instead of exponent	Dottie Pike was tasting tailgate brownies in the Parking Field. Her scorecard has the time on it.
10	B. $y = 500(20)^x$	used 20 instead of 1.2	Not one tarp on the fairgrounds has been moved since morning. The dew on them is undisturbed.
10	D. $y = 500(0.2)^x$	used the rate instead of 1+rate	Mr. Corbin was untangling a hay bale pile in the Hay Maze. Three kids cheered him on.